supremo

$$\phi_i(t+1) = (\alpha - k)\phi_i(t) + \beta \sum_{j \in N(i)} w_{ij}\phi_j(t) + \gamma G_i(t) + \mu M_i(t) +$$

$$K\phi_i^{(ea)}(t) + \rho H_i\{\phi, t\} + \sigma(s(t), t)\eta_i(t)$$

- let $D_i(t) = \gamma G_i(t) + \mu M_i(t)$

applying kernel to $D_i(t)$ say some λ is the distance spread:-

then the equation becomes:

$$\phi_i(t+1) = (\alpha - k)\phi_i(t) + \beta \sum_{j \in N} w_{ij}\phi_j(t) + \lambda \sum_{j \in N(i)} w_{ij} D_j(t) +$$

$$\rho H_i(\phi_i(t)) + \sigma_i(t)\eta_i(t).$$

$\alpha \rightarrow$ big correction

$$\phi_i(t+1) = (\alpha - \lambda)\phi_i(t) + \lambda\phi_i^{(ea)}(t) + \beta(L\phi(t))_i +$$
$$K(LD(t))_i + \varepsilon((J-I)\phi(t))_i + \rho H_i(\phi_i(t)) +$$
$$\sigma_i(t)\eta_i(t)$$

L - graph laplacian $D = aG(t) + bM(t)$.

but $\varepsilon((J-I)\phi(t))_i \rightarrow$ uniform makes sense but we need good picture

$$\phi_i(t+1) = (\alpha - \lambda)\phi_i(t) + \lambda\phi^{(ea)}(t) + \beta(L\phi(t))_i + K(L(D(t)))_i +$$

$$\varepsilon \sum_{j \in N(i)} e^{-d_{ij}/\xi} \phi_j(t) + \rho H_i(\phi_i(t)) + \sigma_i(t)\eta_i(t)$$

supremo

$d_{ij} \rightarrow$ shortest-path from (i, j)

$\Rightarrow$ final equation.

$$\phi_i(t+1) = (\alpha-\lambda)\phi_i(t) + \lambda\phi^{(eq)}(t) + \beta(L\phi(t))_i + K(LD(t))_i +$$

$$\varepsilon \sum_{j \in N(i)} \omega_{ij}\phi_j(t) + \rho H_i(\phi_i(t)) + \sigma_i(t)\eta_i(t).$$

where $\rightarrow$ $a G_i(t) + \beta M_i(t) = D(t)_i$

$\Rightarrow$ $\sigma_i(t)\eta_i(t) = \eta_i(t)\sigma(G^f(t) + M^f(t))$

$$\phi_i(t+1) = (\alpha-\lambda)\phi_i(t) + \lambda\phi^{(eq)}(t) + \beta(L\phi(t))_i + K(LD(t))_i$$

$$+ \varepsilon \sum_{j \in N(i)} \omega_{ij}\phi_j(t) + \rho H_i(\phi_i(t)) + \sigma_i(t)\eta_i(t).$$

$\Rightarrow$

$$\phi_i(t+1) = (\alpha-\lambda)\phi_i(t) + \lambda\phi^{(eq)}(t) + \beta(L\phi(t))_i + K(LD(t))_i +$$

$$\varepsilon \sum_{j \in N(i)} \omega_{ij}\phi_j(t) + \rho H_i(\phi_i(t)) + \eta_i(t)\sigma_i(t).$$

memory and relaxation lag $\rightarrow$ $(\alpha-\lambda)\phi_i(t)$.

final relaxation state $\rightarrow$ $\lambda\phi^{(eq)}(t)$

Correlation term $\rightarrow$ $\beta(L\phi(t))_i$ (Laplacian on graph)

Gate/Measurement

Uniform Disturbance Correlation $\rightarrow$ $K(LD(t))_i =$

$K(L\{aG_i(t) + bM_i(t)\})_i$

supremo

$+ \varepsilon \sum_{j \in N(i)} e^{-d_{ij}/\ell} \phi_j(t) \Rightarrow$ exponential fall off

$d_{ij} \rightarrow$ shortest distance from $i-j$

$\ell = 3/4$

o.r.

$+ \rho H_i(\phi_i(t)) \rightarrow$ non-linear noise factor.

$\tanh(\phi_i(t))$

$+ \eta_i(t)\sigma_i(t) \rightarrow$ stochaistic noise jumps.

the original equation:- $H_i\{\phi_i(t)\} \Rightarrow x/1+x^2$.

$$\phi_i(t+1) = (\alpha)\phi_i(t) + \beta \sum_{j \in N(i)} w_{ij}\phi_j(t) + \gamma G_i(t) + \mu M_i(t) + \rho H_i\{\phi_i(t)\} + \sigma_i(t)\eta_i(t)$$

→ after adding adiamplitude dampening terms:

$$\phi_i(t+1) = (\alpha-\lambda)\phi_i(t) + \beta \sum_{j \in N(i)} w_{ij}\phi_j(t) + \gamma G_i(t) + \mu M_i(t) + \rho H_i\{\phi_i(t)\} + \sigma_i(t)\eta_i(t) + \lambda \phi_i^{(e)}(t)$$

→ Coupling gate and measurement burst into Disturbance → $D_i(t) = aG_i(t) + bM_i(t)$

$$\rightarrow \phi_i(t+1) = (\alpha-\lambda)\phi_i(t) + \lambda\phi_i^{(ea)}(t) + \beta \sum_{j \in N(i)} w_{ij}\phi_j(t) + k \sum_{j \in N(i)} w_{ij} D_i(t) + \rho H_i\{\phi_i(t)\} + \sigma_i(t)\eta_i(t)$$

→ Replacing w_{ij} → Laplacian term and adding a uniform exponential term ε.

$$\phi_i(t+1) = (\alpha-\lambda)\phi_i(t) + \lambda\phi_i^{(ea)}(t) + \beta(L\phi(t))_i + k(LD(t))_i + \varepsilon \sum_{j \in N(i)} w_{ij}\phi_j(t) + \rho H_i\{\phi_i(t)\} + \eta_i(t)\sigma_i(t).$$

$$\phi_i = \tanh(\phi_i)$$

Final Equation:

$$\phi_i(t+1) = (\alpha-\lambda)\phi_i(t) + \beta(L\phi_i(t)) + k(LD_i(t)) + \varepsilon\sum_{j\in N(i)} w_{ij}\phi_j(t) + \frac{\rho\phi_i(t)}{1+\phi_i(t)^2} + \sigma(G_i^{(t)}(t) + M_i^{(t)}(t))\eta_i(t)$$

and $\lambda\phi^{(eq)}(T) = 0$

let the given feature map be.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} (\alpha-\lambda)\phi_i(t) \\ \beta(L\phi(t))_i \\ k(LD(t))_i \\ \varepsilon\sum w_{ij}\phi_i(t) \\ \rho\phi_i(t)/1+\phi_i(t)^2 \\ \sigma(G+M)\eta_i(t) \end{bmatrix} = \Phi(t+1) \rightarrow \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix}$$

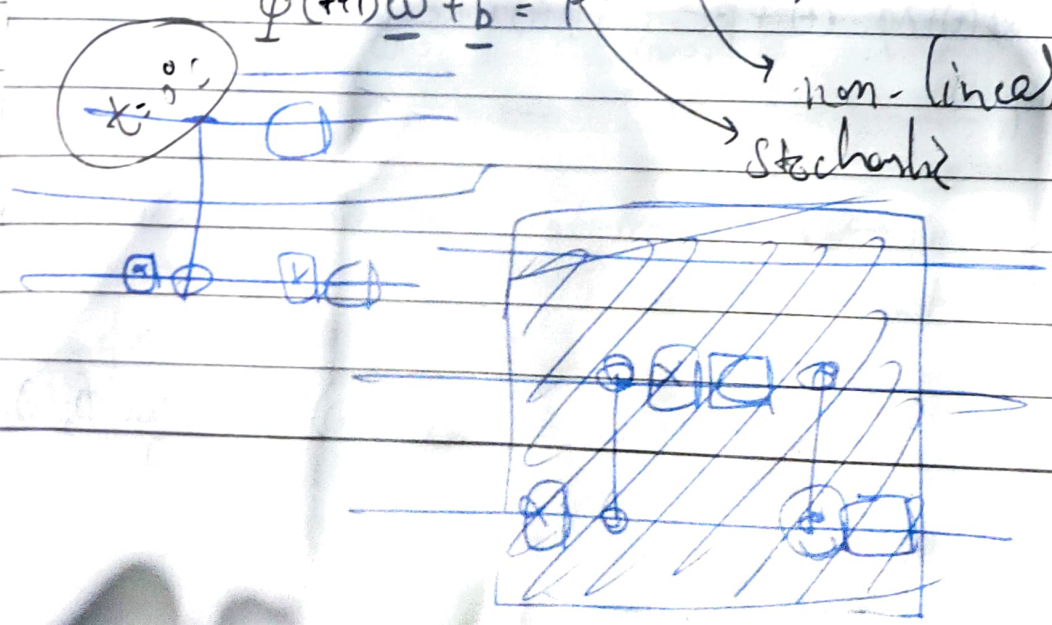
mem

correlation (qubit)

mixed gate and mem uremen t

Baking Noise

non-linear

Stochastic

$$\Phi(t+1)\underline{w} + \underline{b} = P^{(k)}.$$